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May 7, 2026

Editorial Office
Bioinformatics
Oxford University Press
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Dear Editor,

We are pleased to submit our manuscript entitled:

"Large-scale paired prediction of GPCR–G protein coupling specificity with protein language models and topology-aware features"

for consideration as an **Original Paper** in *Bioinformatics*.

Summary

Predicting GPCR–G protein coupling specificity is a long-standing challenge in computational biology, with direct implications for orphan receptor deorphanization and rational drug design. In this study, we present a systematic computational framework that reformulates the problem as paired (GPCR, G protein family) binary classification, using a curated dataset of 1,647 experimentally annotated pairs spanning 431 GPCRs and 4 G protein families.

Our main contributions are:

1. **Systematic architecture comparison:** We benchmark SVM (RBF), cross-attention deep networks, multi-task learning, MLP, Random Forest, and XGBoost under identical conditions, using ESM-2 embeddings at two scales (8M, 650M).
2. **Topology-aware ICL features:** Using UniProt transmembrane annotations, we extract ICL2/3 local features and demonstrate a critical

dimension alignment requirement: local features must match the global embedding dimension.

3. **Negative result with practical impact:** 38-dimensional AlphaFold structural descriptors provide no incremental benefit beyond ESM-2 650M + ICL, suggesting that high-capacity PLMs implicitly encode sufficient structural information—a finding with practical implications for future work.
4. **Multi-task analysis:** We show that removing G protein identity information degrades AUC from 0.862 to 0.802, confirming the paired formulation is essential.

The best-performing model (cross-attention + 650M ESM-2 + ICL features) achieves an AUC of 0.8619 ± 0.0249 under strict cluster-aware cross-validation, significantly outperforming SVM (0.8331).

Why Bioinformatics?

We believe this work is well-suited for *Bioinformatics* because it provides (i) a rigorous benchmarking framework for GPCR–G protein coupling prediction, (ii) generalizable methodological insights (dimension alignment, AlphaFold redundancy) that apply beyond this specific task, and (iii) fully reproducible code and data.

Data and Code Availability

All code, data, and documentation are publicly available at `https://github.com/nblvguohao/topology-aware-gpcr-coupling` and archived on Zenodo.

Conflict of Interest

The authors declare no competing interests.

Thank you for considering our submission. We look forward to your response.

Qingyong Wang and Lichuan Gu

Sincerely,

Qingyong Wang and Lichuan Gu

On behalf of all co-authors